

Basic Data Structures & Algorithms

Questions and Answers for Beginners

Topics: Arrays, Strings, Linked Lists, Stack, Queue, Searching, Sorting, Trees, Graphs and Complexity.

1. What is DSA?

DSA stands for Data Structures and Algorithms. Data structures organize and store data, while algorithms are step-by-step procedures used to solve problems efficiently.

2. What is a Data Structure?

A data structure is a method of organizing and storing data so that it can be accessed and modified efficiently. Examples include arrays, linked lists, stacks, queues, trees and graphs.

3. What is an Algorithm?

An algorithm is a finite sequence of clear steps used to solve a particular problem or perform a computation.

4. What is an Array?

An array is a collection of elements stored in a contiguous memory location. Elements are accessed using an index. In most programming languages, array indexing starts from 0.

5. What is a String?

A string is a sequence of characters. For example, "DATA" contains four characters: D, A, T and A.

6. What is a Linked List?

A linked list is a linear data structure made of nodes. Each node contains data and a reference (link) to another node.

7. What is a Stack?

A stack is a linear data structure that follows LIFO (Last In, First Out). The last element inserted is the first element removed. Common operations are push and pop.

8. What is a Queue?

A queue is a linear data structure that follows FIFO (First In, First Out). The first element inserted is the first element removed. Common operations are enqueue and dequeue.

9. What is Linear Search?

Linear search checks elements one by one until the required element is found or the list ends. Its worst-case time complexity is $O(n)$.

10. What is Binary Search?

Binary search works on a sorted array by repeatedly dividing the search range into two halves. Its time complexity is $O(\log n)$.

11. What is Sorting?

Sorting means arranging data in a particular order, such as ascending or descending order. Examples include Bubble Sort, Selection Sort, Insertion Sort, Merge Sort and Quick Sort.

12. What is Bubble Sort?

Bubble Sort repeatedly compares adjacent elements and swaps them when they are in the wrong order. Its average and worst-case time complexity is $O(n^2)$.

13. What is Recursion?

Recursion is a technique in which a function calls itself to solve smaller versions of the same problem. A recursive solution must have a base condition to stop the calls.

14. What is a Tree?

A tree is a non-linear hierarchical data structure consisting of nodes connected by edges. A tree has a root node and may contain child nodes.

15. What is a Binary Tree?

A binary tree is a tree in which each node can have at most two children, commonly called the left child and right child.

16. What is a Graph?

A graph is a non-linear data structure consisting of vertices (nodes) and edges that represent relationships between the vertices.

17. What is Time Complexity?

Time complexity describes how the running time of an algorithm grows as the input size increases. Common complexities include $O(1)$, $O(\log n)$, $O(n)$, $O(n \log n)$ and $O(n^2)$.

18. What is Space Complexity?

Space complexity describes how much additional memory an algorithm needs as the input size increases.

19. Difference between Stack and Queue?

A stack follows LIFO, while a queue follows FIFO. Stack operations are generally performed at one end, whereas a queue inserts at the rear and removes from the front.

20. What is Big-O Notation?

Big-O notation is used to describe the upper-bound growth rate of an algorithm's time or space requirements as the input size becomes large.

Basic DSA Programming Questions

21. Find the largest element in an array.

```
arr = [10, 25, 7, 40, 15]
largest = arr[0]

for x in arr:
    if x > largest:
        largest = x

print("Largest:", largest)
```

Output: Largest: 40

22. Reverse an array.

```
arr = [1, 2, 3, 4, 5]
arr.reverse()
print(arr)
```

Output: [5, 4, 3, 2, 1]

23. Perform linear search.

```
arr = [10, 20, 30, 40, 50]
target = 30

for i in range(len(arr)):
    if arr[i] == target:
        print("Found at index:", i)
        break
    else:
        print("Not Found")
```

Output: Found at index: 2

24. Count the number of elements in a list without using len().

```
arr = [5, 10, 15, 20]
count = 0

for x in arr:
    count += 1

print("Count:", count)
```

Output: Count: 4

25. Check whether a string is a palindrome.

```
s = "madam"

if s == s[::-1]:
    print("Palindrome")
else:
    print("Not Palindrome")
```

Output: Palindrome

Quick Complexity Reference

Operation / Algorithm	Best / Typical	Worst
Array access	$O(1)$	$O(1)$
Linear Search	$O(1)$	$O(n)$
Binary Search	$O(1)$	$O(\log n)$
Bubble Sort	$O(n)$	$O(n^2)$
Merge Sort	$O(n \log n)$	$O(n \log n)$
Quick Sort	$O(n \log n)$	$O(n^2)$

Note: These questions are designed for beginners and can be used for basic DSA practice, viva preparation, or interview revision.